



VP AIR TEST NEET (2025-26)

PART TEST - 07

DROPPER

DURATION: 180 Minutes

Date: 21-12-2025

M. MARKS: 720

ANSWER KEY

1. (3)	46. (2)	91. (2)	136. (3)
2. (2)	47. (2)	92. (3)	137. (1)
3. (4)	48. (1)	93. (2)	138. (3)
4. (1)	49. (1)	94. (1)	139. (1)
5. (3)	50. (2)	95. (4)	140. (4)
6. (2)	51. (1)	96. (2)	141. (2)
7. (1)	52. (2)	97. (3)	142. (3)
8. (1)	53. (2)	98. (3)	143. (4)
9. (1)	54. (3)	99. (2)	144. (1)
10. (2)	55. (1)	100. (3)	145. (3)
11. (1)	56. (4)	101. (1)	146. (1)
12. (3)	57. (2)	102. (4)	147. (1)
13. (4)	58. (2)	103. (4)	148. (4)
14. (2)	59. (4)	104. (1)	149. (3)
15. (4)	60. (3)	105. (2)	150. (1)
16. (4)	61. (2)	106. (2)	151. (4)
17. (3)	62. (2)	107. (3)	152. (1)
18. (4)	63. (4)	108. (3)	153. (2)
19. (2)	64. (1)	109. (2)	154. (4)
20. (1)	65. (2)	110. (4)	155. (1)
21. (2)	66. (2)	111. (2)	156. (1)
22. (3)	67. (2)	112. (2)	157. (2)
23. (2)	68. (1)	113. (1)	158. (3)
24. (4)	69. (3)	114. (4)	159. (3)
25. (2)	70. (3)	115. (3)	160. (4)
26. (1)	71. (4)	116. (4)	161. (3)
27. (1)	72. (4)	117. (2)	162. (4)
28. (4)	73. (3)	118. (1)	163. (2)
29. (3)	74. (1)	119. (4)	164. (1)
30. (4)	75. (4)	120. (3)	165. (2)
31. (3)	76. (2)	121. (1)	166. (4)
32. (3)	77. (2)	122. (2)	167. (2)
33. (2)	78. (2)	123. (3)	168. (3)
34. (4)	79. (4)	124. (4)	169. (3)
35. (4)	80. (2)	125. (2)	170. (4)
36. (1)	81. (1)	126. (1)	171. (2)
37. (1)	82. (4)	127. (3)	172. (2)
38. (3)	83. (1)	128. (1)	173. (2)
39. (3)	84. (4)	129. (1)	174. (3)
40. (4)	85. (1)	130. (1)	175. (4)
41. (1)	86. (1)	131. (3)	176. (1)
42. (2)	87. (3)	132. (4)	177. (3)
43. (3)	88. (3)	133. (2)	178. (2)
44. (3)	89. (2)	134. (1)	179. (3)
45. (2)	90. (2)	135. (3)	180. (4)

HINTS & SOLUTIONS

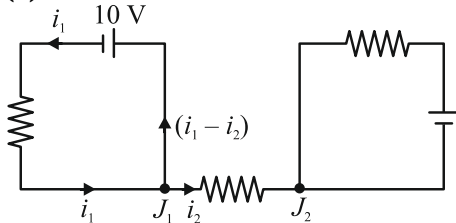
1. (3)

$$U_{total} = U_1 + U^2 + U_{int}$$

$$U_{total} = \frac{kQ_1^2}{2R_1} + \frac{kQ_2^2}{2R_2} + \frac{kQ_1Q_2}{r}$$

(NCERT Ed. 2025-26, 12th, Page No. 55)

2. (2)



No current can go through 2Ω resistor because according to junction law, the incoming current should be equal to outgoing current.

At junction J_1 :

Current going to battery $= (i_1 - i_2)$

Current coming out of battery $= i_1$

$$i_1 = (i_1 - i_2)$$

$$\text{Hence, } i_2 = 0$$

(NCERT Ed. 2025-26, 12th, Page No. 97)

3. (4)

Equilibrium Condition: Net torques must be zero.

$$\tau_{electric} = \tau_{external}$$

$$pE \sin \theta = \frac{pE}{2} \Rightarrow \sin \theta = \frac{1}{2}$$

In the range 0° to 180° , the solutions are $\theta = 30^\circ$ and $\theta = 150^\circ$.

(NCERT Ed. 2025-26, 12th, Page No. 60)

4. (1)

Due to the principle of superposition, potential at O

$$V = \frac{1}{4\pi\epsilon_0} \times \frac{28 \times 10^{-6}}{0.4} = 9 \times 10^9 \times \frac{28 \times 10^{-6}}{0.4}$$

$$= 63 \times 10^4 \text{ volt}$$

(NCERT Ed. 2025-26, 12th, Page No. 51)

5. (3)

Range of voltmeter $= (10 \times 10^{-3}) (90 + 910) = 10 \text{ V}$

$$\text{No. of divisions} = \frac{10}{0.1} = 100$$

(NCERT Ed. 2025-26, 12th, Page No. 96)

6. (2)

$$V = \frac{1000}{x} + \frac{1500}{x^2} + \frac{500}{x^3}$$

$$E = -\frac{\partial V}{\partial x} \hat{i} - \frac{\partial V}{\partial y} \hat{j} - \frac{\partial V}{\partial z} \hat{k}$$

$$\frac{\partial V}{\partial x} = -\frac{1000}{x^2} - \frac{3000}{x^3} - \frac{1500}{x^4}$$

$$\frac{\partial V}{\partial y} = 0 \text{ \& } \frac{\partial V}{\partial z} = 0$$

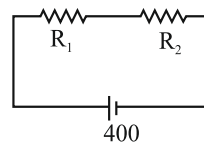
$$\therefore E(x, 0, 0) = -\left(\frac{1000}{x^2} + \frac{3000}{x^3} + \frac{1500}{x^4}\right) \hat{i} \text{ V/m}$$

$$E(1, 0, 0) = -(1000 + 3000 + 1500) \hat{i} \text{ V/m}$$

$$= -5500 \hat{i} \text{ V/m}$$

(NCERT Ed. 2025-26, 12th, Page No. 55)

7. (1)



$$R_1 = \frac{220^2}{25}, R_2 = \frac{220^2}{100}$$

$$R_1 = 4R_2$$

$$V_1 = \frac{400 \times 4R}{4R + R} = 320$$

$$V_2 = \frac{400 \times R}{5R} = 80$$

For 25 watt,

$320 > 220$, so bulb will fuse.

(NCERT Ed. 2025-26, 12th, Page No. 93)

8. (1)

$$V = \frac{kq}{r}, \text{ on combination } q' = nq \text{ and}$$

$$r' = \left(\frac{n \left(\frac{4}{3} \pi r^3 \right)}{\frac{4}{3} \pi} \right)^{\frac{1}{3}} = n^{\frac{1}{3}} r$$

$$V' = \frac{k(nq)}{n^{\frac{1}{3}} r} = n^{\frac{2}{3}} \frac{kq}{r} = n^{\frac{2}{3}} V$$

(NCERT Ed. 2025-26, 12th, Page No. 48)

9. (1)

$$Q = t - t^2$$

$$i = \frac{dQ}{dt} = 1 - 2t \text{ (decreases linearly with time in}$$

$$\text{interval } t = 0 \text{ to } t = \frac{1}{2} \text{ s)}$$

at $t = 0$,

$$i = 1 \text{ Amp}$$

for $i = 0$

$$1 - 2t = 0$$

$$\Rightarrow t = \frac{1}{2} \text{ s}$$

(NCERT Ed. 2025-26, 12th, Page No. 81)

10. (2)

Gain in K.E. = Loss in P.E.

Moment of Inertia (I):

$$I = \frac{ml^2}{12} (\text{rod}) + 2 \times m \left(\frac{l}{2} \right)^2 (\text{masses})$$

$$= \frac{7ml^2}{12}$$

Energy Balance:

$$\frac{1}{2} I \omega^2 = pE(1 - \cos 53^\circ)$$

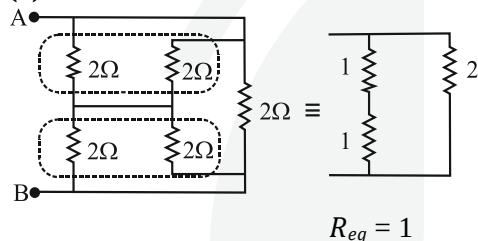
$$\frac{1}{2} \left(\frac{7ml^2}{12} \right) \omega^2 = (ql)E(1 - 0.6)$$

$$\frac{7ml^2}{24} \omega^2 = 0.4qlE$$

$$\omega = \sqrt{\frac{48qE}{35ml}}$$

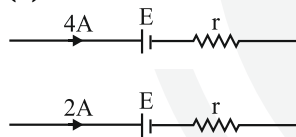
(NCERT Ed. 2025-26, 12th, Page No. 60)

11. (1)



(NCERT Ed. 2025-26, 12th, Page No. 100)

12. (3)



$$E + 4r = 12$$

$$E - 2r = 9$$

$$\Rightarrow E = 10 \text{ V and } r = 0.5$$

(NCERT Ed. 2025-26, 12th, Page No. 94)

13. (4)

$$U = \frac{kq_1q_2}{r}$$

so, PE is maximum when it is positive and r is minimum i.e. closest to $+q$ charge at point D .

(NCERT Ed. 2025-26, 12th, Page No. 54)

14. (2)

$$P_1 = \frac{V^2 A}{\rho \ell}$$

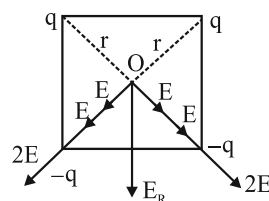
$$P_2 = \frac{V^2 A}{\rho(0.9\ell)} = \frac{(1.1)V^2 A}{\rho \ell}$$

(NCERT Ed. 2025-26, 12th, Page No. 93)

15. (4)

$$V_0 = \frac{kq}{r} + \frac{kq}{r} - \frac{kq}{r} - \frac{kq}{r} = 0$$

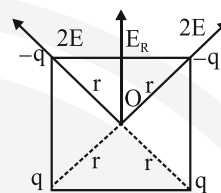
$$E_R = 2(2E) \cos 45(-\hat{j}) = 2\sqrt{2} \frac{kq}{r^2} (-\hat{j})$$



After interchanging

$$V_0 = \frac{kq}{r} + \frac{kq}{r} - \frac{kq}{r} - \frac{kq}{r} = 0$$

$$E_R = 2(2E) \cos 45(+\hat{j}) = 2\sqrt{2} \frac{kq}{r^2} (+\hat{j})$$



Direction of E changes.

(NCERT Ed. 2025-26, 12th, Page No. 51)

16. (4)

$$C = \frac{A\epsilon_0}{d}$$

(NCERT Ed. 2025-26, 12th, Page No. 68)

17. (3)

$$C_2 = 2C_1$$

$$Q_2 = 2Q_1$$

$$E_1 = \frac{V}{d} \quad E_2 = \frac{V}{d/2} = 2\left(\frac{V}{d}\right)$$

$$F_1 = E_1 Q_1 \quad F_2 = E_2 Q_2$$

$$\Rightarrow F_2 = 4F_1$$

(NCERT Ed. 2025-26, 12th, Page No. 74)

18. (4)

$$I = neAv_d$$

Here I and n is constant

$$v_d \propto \frac{1}{A}$$

$$\frac{v'}{v} = \frac{\frac{\pi d^2}{4}}{\frac{\pi d'^2}{4}} = \frac{d^2}{d'^2}$$

$$v' = 4v$$

(NCERT Ed. 2025-26, 12th, Page No. 86)

19. (2)

$$q_i = CV$$

$$q_f = 2CV$$

$$\Delta q = q_f - q_i = q_f - (-q_i) = 3CV$$

$$U_i = \frac{1}{2} CV^2$$

$$U_f = \frac{1}{2} \times \frac{(2CV)^2}{C}$$

$$W_{\text{battery}} = \Delta H + \Delta U$$

$$(3CV) \times 2V = 2CV^2 - \frac{CV^2}{2} + H$$

$$\text{Ratio} = \frac{H}{U_f} = 2.25$$

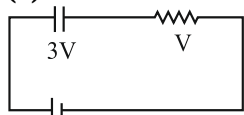
(NCERT Ed. 2025-26, 12th, Page No. 74)

20. (1)

As capacity of system increases so the charge supplied by the battery increases.

(NCERT Ed. 2025-26, 12th, Page No. 70)

21. (2)



$4V = 12 \Rightarrow V = 3$ volt
and $3V = 9$ volt

$$q = 36 \mu\text{C}$$

$$RC = 2.5 \times 10^6 \times 4 \times 10^{-6}$$

$$= 10 \text{ sec}$$

$$\therefore q = CV(1 - e^{-t/RC})$$

$$36 \mu\text{C} = 4 \times 12(1 - e^{-t/10})$$

$$0.75 = (1 - e^{-t/10})$$

$$e^{-t/10} = 0.25$$

$$t = 13.86 \text{ s}$$

(NCERT Ed. 2025-26, 12th, Page No. 97)

22. (3)

$$I = \frac{Q}{t} = \frac{1.6 \times 10^{-19} \times 10^{20}}{1} = 16 \text{ A}$$

(NCERT Ed. 2025-26, 12th, Page No. 81)

23. (2)

The average velocity is zero, in case of no current due to random motion.

(NCERT Ed. 2025-26, 12th, Page No. 85)

24. (4)

On decreasing temperature resistance of metal will decrease and that of semiconductor will increase.

(NCERT Ed. 2025-26, 12th, Page No. 90)

25. (2)

In the loop ABCD

$$\varepsilon_1 - i_1 r_1 - (i_1 + i_2)R = 0 \text{ is correct}$$

The other equations are wrong

(NCERT Ed. 2025-26, 12th, Page No. 97)

26. (1)

$$V = \frac{Kp \cos \theta}{r^2} = \frac{K \vec{p} \cdot \vec{r}}{r^3}$$

(NCERT Ed. 2025-26, 12th, Page No. 51)

27. (1)

As the capacitor is isolated, the charge on the capacitor will not change.

(NCERT Ed. 2025-26, 12th, Page No. 74)

28. (4)

$$\therefore R = \frac{V^2}{P} = \frac{(220)^2}{100} = 484 \Omega$$

$$\text{Resistance of half piece} = \frac{484}{2} = 242 \Omega$$

$$\text{Resistance of pieces when connected in parallel}$$

$$= \frac{242}{2} = 121 \Omega$$

Energy liberated/second

$$P = \frac{V^2}{R} = \frac{220 \times 220}{121} = 400 \text{ W}$$

(NCERT Ed. 2025-26, 12th, Page No. 93)

29. (3)

$$\text{A. } \frac{Kq}{r}$$

$$\text{B. } \frac{Kq}{R}$$

$$\text{C. } \frac{Kq}{2R^3}(3R^2 - r^2)$$

(NCERT Ed. 2025-26, 12th, Page No. 48)

30. (4)

$$\text{For long wire } E = \frac{2k\lambda}{r}$$

$$\int dV = - \int_a^b \vec{E} \cdot d\vec{r}$$

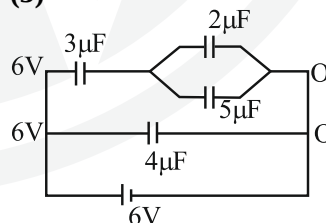
$$\Delta V = - \int_a^b \frac{2K\lambda}{r} dr = -2K\lambda(\ln r)_a^b$$

$$\Delta V = -2K\lambda \log_e \left(\frac{b}{a} \right)$$

$$\Rightarrow \Delta V \propto \log_e \left(\frac{b}{a} \right)$$

(NCERT Ed. 2025-26, 12th, Page No. 55)

31. (3)



$$Q_{\text{upper}} = \left(\frac{21 \times 6}{10} \right) \mu\text{C}$$

$$\frac{Q_{5\mu\text{F}}}{Q_{4\mu\text{F}}} = \frac{\left(5 \times \frac{18}{10} \right)}{4 \times 6} = \frac{3}{8}$$

(NCERT Ed. 2025-26, 12th, Page No. 71)

32. (3)

$$U = -\vec{p} \cdot \vec{E}$$

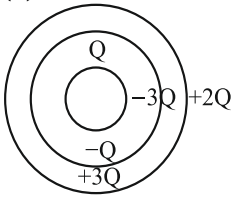
$$= -PE \cos \theta$$

For unstable equilibrium, U should be maximum

$$\text{For } U_{\text{max}} \Rightarrow \theta = 180^\circ$$

(NCERT Ed. 2025-26, 12th, Page No. 60)

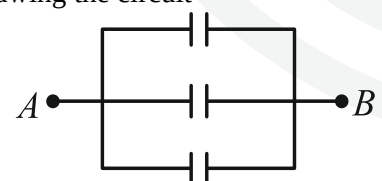
33. (2)
 $\frac{5}{\ell_1} = \frac{R}{100 - \ell_1} \Rightarrow 500 - 5\ell_1 = R\ell_1 \dots\dots(i)$
 $\frac{5}{1.6\ell_1} = \frac{R}{100 - 1.6\ell_1} \Rightarrow 500 - 8\ell_1 = 0.8R\ell_1 \dots\dots(ii)$
 From (i) and (ii)
 $500 - 8\ell_1 = 0.8(500 - 5\ell_1)$
 $\ell_1 = 25 \text{ cm}$
 $R = 15 \Omega$
 (NCERT Ed. 2025-26, 12th, Page No. 101)

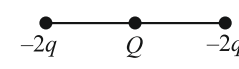
34. (4)

 (NCERT Ed. 2025-26, 12th, Page No. 62)

35. (4)
 $V_A - (2 \times 6) - 12 - (2 \times 9) + 4 - (2 \times 5) = V_B$
 $V_A - V_B = 48 \text{ V.}$
 (NCERT Ed. 2025-26, 12th, Page No. 97)

36. (1)
 $\Rightarrow \Delta V = \frac{W}{q} = \frac{3.2 \times 10^{-10}}{1.6 \times 10^{-19}}$
 $= 2 \times 10^9 \text{ V}$
 (NCERT Ed. 2025-26, 12th, Page No. 47)

37. (1)
 All the statements are correct.
 (NCERT Ed. 2025-26, 12th, Page No. 62)

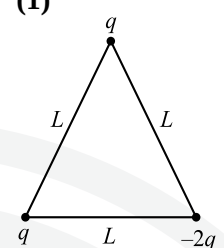
38. (3)
 Redrawing the circuit

 $C_{eq} = 3C = \frac{3\epsilon_0 A}{d}$
 (NCERT Ed. 2025-26, 12th, Page No. 71)

39. (3)

 $U = \frac{-2kQq}{r} \times 2 + \frac{4kq^2}{2r} = 0$
 $\Rightarrow \frac{-4kQq}{r} + \frac{2kq^2}{r} = 0$
 $q = 2Q$

$$\Rightarrow \frac{Q}{q} = \frac{1}{2}$$

(NCERT Ed. 2025-26, 12th, Page No. 55)

40. (4)
 $C = 4\pi\epsilon_0 R$
 $R = \left(\frac{1}{4\pi\epsilon_0} \right) \times C = kC = 9 \times 10^9 \times 10 \times 10^{-12}$
 $R = 90 \times 10^{-3} \text{ m} = 9 \text{ cm.}$
 (NCERT Ed. 2025-26, 12th, Page No. 67)

41. (1)

 $U = \frac{k(q)(-2q)}{L} + \frac{k(q)(q)}{L} + \frac{k(q)(-2q)}{L}$
 $U = \frac{-3kq^2}{L}$
 $U_\infty = 0$
 $W_{ext} = U_\infty - U = \frac{3kq^2}{L}$
 (NCERT Ed. 2025-26, 12th, Page No. 56)

42. (2)
 $R = \frac{\rho l^2}{V}, R' = \frac{\rho(0.8l)^2}{V} = 0.64R$
 Percentage decrease = $\frac{R - R'}{R} \times 100$
 $= \left(\frac{R - 0.64R}{R} \right) \times 100 = 36\%$
 (NCERT Ed. 2025-26, 12th, Page No. 84)

43. (3)
 $Q_1 = Q_2 + Q_3$
 $V_2 = V_3$
 $V = V_1 + V_2$
 (NCERT Ed. 2025-26, 12th, Page No. 71)

44. (3)
 In presence of conservative force work done in moving on a closed path is zero, $\therefore W_{ext} = 0$.
 (NCERT Ed. 2025-26, 12th, Page No. 54)

45. (2)
 $E_{eq} = \frac{E \times r + E \times r}{r + r} = E$
 $r_{eq} = \frac{r \times r}{r + r} = \frac{r}{2}$
 (NCERT Ed. 2025-26, 12th, Page No. 96)

46. (2)

It is an example of electrophilic addition reaction as the alkene abstracts H^+ ion to form carbocation in the first step.

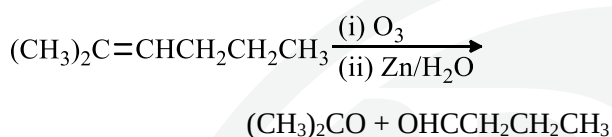
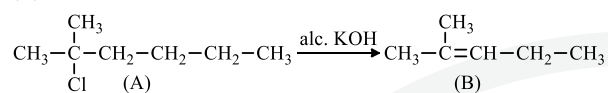
(NCERT Ed. 2025-26 11th Part-II Page No. 313)

47. (2)

Less hindered haloalkane follows S_N2 mechanism.

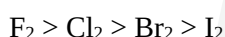
(NCERT Ed. 2025-26 12th Part-II Page No. 166)

48. (1)

(NCERT Ed. 2025-26 11th Part-II Page No. 313)

49. (1)

The rate of reaction of alkanes with halogens is

(NCERT Ed. 2025-26 11th Part-II Page No. 303)

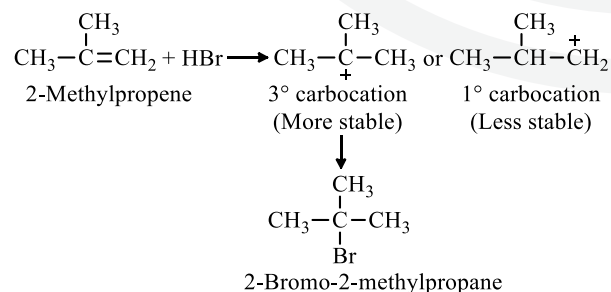
50. (2)

Number of alkyl chloride obtained on monochlorination of the given alkanes respectively are:

(A) 3, (B) 3, (C) 5, (D) 1, (E) 5, (F) 4, (G) 2, (H) 4

(NCERT Ed. 2025-26 11th Part-II Page No. 302)

51. (1)

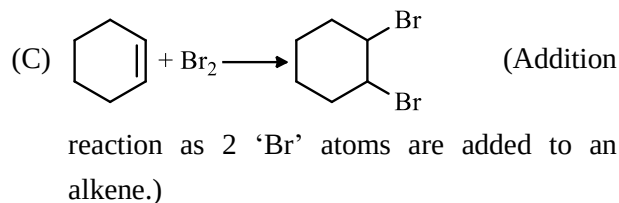
(NCERT Ed. 2025-26 11th Part-II Page No. 311)

52. (2)



It is an elimination reaction.

(B) Reaction involves substitution of 'Br' with 'OH'. So, it is a substitution reaction.

(NCERT Ed. 2025-26 12th Part-II Page No. 171)

53. (2)

$-\text{NH}_2$ group activates the benzene ring and hence the incoming electrophile attaches at ortho and para position.

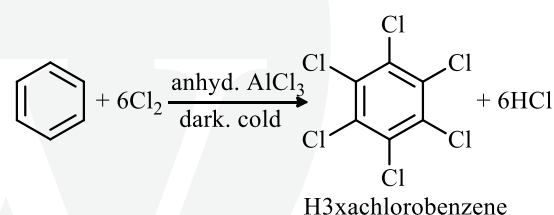
(NCERT Ed. 2025-26 11th Part-II Page No. 325)

54. (3)

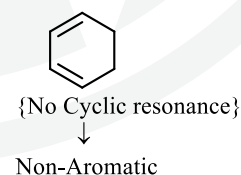
Of all the conformations of ethane, the staggered form has the least torsional strain and the eclipsed form has the maximum torsional strain. Therefore, staggered conformation is more stable than the eclipsed conformation.

(NCERT Ed. 2025-26 11th Part-II Page No. 305)

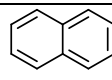
55. (1)

(NCERT Ed. 2025-26 11th Part-II Page No. 324)

56. (4)

(NCERT Ed. 2025-26 11th Part-II Page No. 321)

57. (2)

	Aromatic and has 6π electrons
	Aromatic and has 10π electrons
	Anti-aromatic and has 4π electrons
	Non-aromatic and has 4π electrons

(NCERT Ed. 2025-26 11th Part-II Page No. 321)

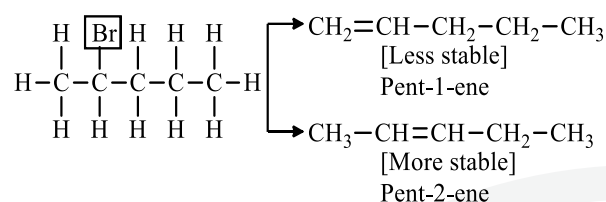
58. (2)

Aryl halides are less reactive towards nucleophilic substitution reaction than alkyl halides because in aryl halides, C-Cl bond acquires a partial double bond character.

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59. (4)

It is a dehydrohalogenation reaction (β -elimination)



Most substituted product will be most stable therefore the most favored.

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60. (3)

- Acetylene is more acidic than neo-pentane and But-1-ene.
- s-character of C-H bond is high in acetylene because it is sp hybridised.
- It is a gas at room temperature.

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61. (2)

Addition of HBr will take place on double bond (other than benzene) with the formation of more stable carbocation.

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62. (2)

The conformation in which the heavier groups are present at maximum possible distances, so that the forces of repulsion get weak, is more stable.

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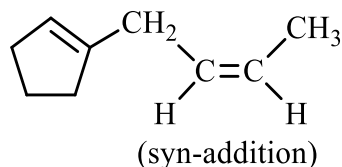
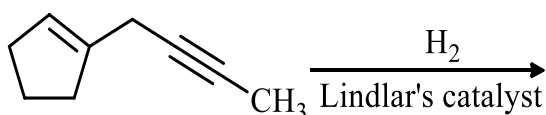
63. (4)

C_1 - C_4 alkanes are gases.

So, C_5 - C_{17} alkanes are liquids.

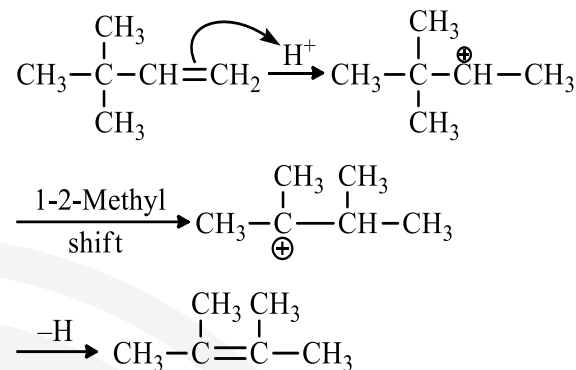
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64. (1)



(NCERT Ed. 2025-26 11th Part-II Page No. 309)

65. (2)



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66. (2)

For these alkenes, stability mainly depends on the degree of substitution and steric strain:

- 1-Butene = monosubstituted (least stable)
- cis-2-Butene = disubstituted, but has steric repulsion (less stable than trans)
- trans-2-Butene = disubstituted and least steric strain (most stable)

So the decreasing order of stability is:

trans-2-Butene (III) > cis-2-Butene (II)
> 1-Butene (I)

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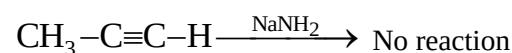
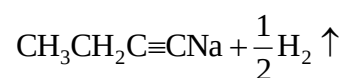
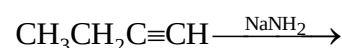
67. (2)

Chloroform is slowly oxidised by air in the presence of light to an extremely poisonous gas, carbonyl chloride, also known as phosgene.

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68. (1)

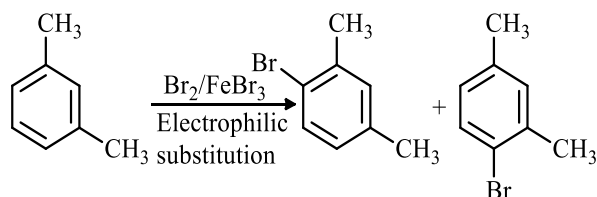
NaNH_2 is used to distinguish between 1-butyne and 2-butyne.



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69. (3)

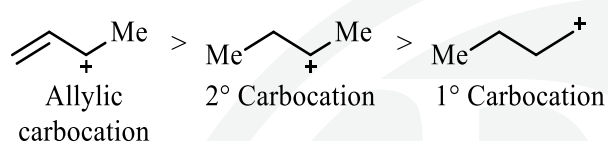
CH₃ is a o/p-directing group, thus electrophilic substitution reaction of toluene



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70. (3)

The reactivity in S_N1 reactions depends upon the stability of the intermediate i.e., carbocation. Since the stability of the carbocation follows the order:

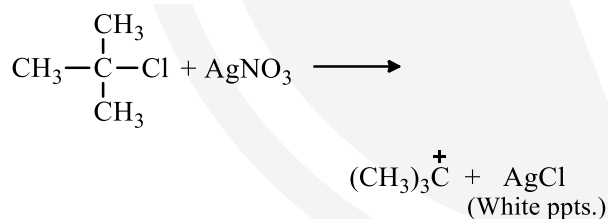


Therefore, S_N1 reactivity of these three bromides follows the order : B > C > A

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71. (4)

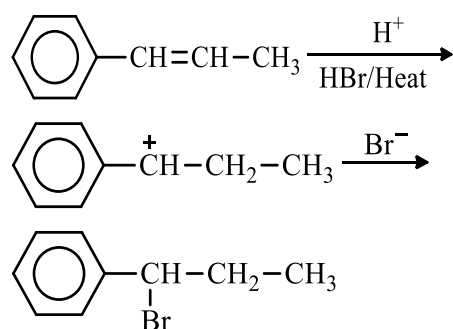
(CH₃)₃CCl gives a precipitate with AgNO₃ solution because it forms stable carbocation. (i.e. tertiary) that readily combines with AgNO₃ to give precipitates of AgCl.



CH₂=CH—Cl forms unstable carbocation. Hence, it does not readily react with AgNO₃.

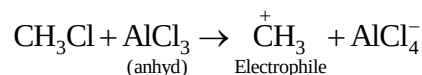
(NCERT Ed. 2025-26 12th Part-II Page No. 171)

72. (4)



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73. (3)



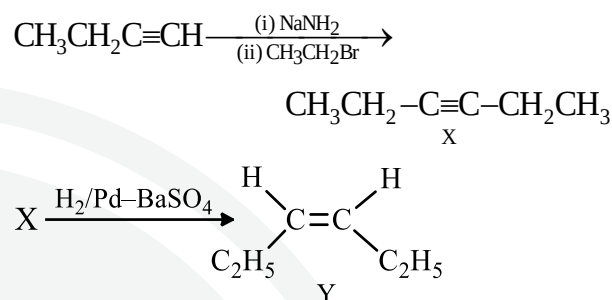
(NCERT Ed. 2025-26 12th Part-II Page No. 171)

74. (1)

- Both I and III undergo S_N1 reaction.
- II undergoes S_N2 reaction.

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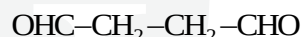
75. (4)



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76. (2)

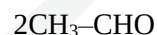
Cyclobutene: Ozonolysis of this four-membered ring with one double bond produces butane-1,4-dial:



Cyclobuta-1,3-diene: Ozonolysis cleaves both double bonds in this four-membered ring, resulting in two molecules of ethane-1,2-dial:



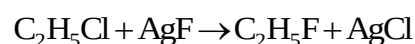
But-2-ene: Ozonolysis of this linear alkene yields two molecules of ethanal:



But-1-ene: Ozonolysis of this linear alkene yields a mixture of methanal (HCHO) and propanal (CH₃—CH₂—CHO)

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77. (2)



This reaction is called Swart's reaction

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78. (2)

In S_N1 reactions, the rate of reaction is dependent only on the concentration of alkyl halide that is, rate = k [RX].

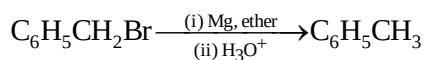
(NCERT Ed. 2025-26 12th Part-II Page No. 171)

79. (4)

H-Cl does not give free radical reaction in the presence of peroxide.

(NCERT Ed. 2025-26 11th Part-II Page No. 311)

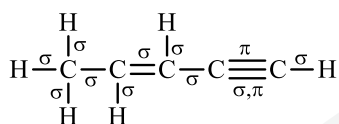
80. (2)



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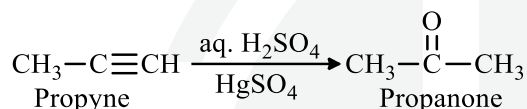
81. (1)

The structure of pent-3-en-1-yne is as below: Here No. of σ -bonds are 10 and no. of π -bonds are 3.



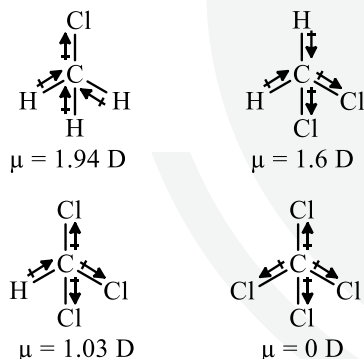
(NCERT Ed. 2025-26 11th Part-II Page No. 300)

82. (4)



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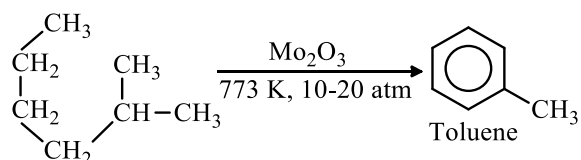
83. (1)



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84. (4)

This is catalytic aromatization of alkanes.



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85. (1)

Isopropyl chloride i.e. $\text{CH}_3-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{Cl}$
(2° - Alkyl halide)

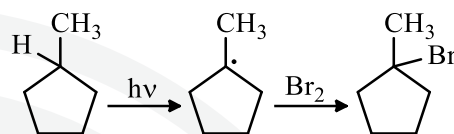
(NCERT Ed. 2025-26 12th Part-II Page No. 160)

86. (1)

$\text{CH}_2=\text{CH}-\text{CH}_2-\text{X}$	Allylic halide
$\text{CH}_2=\text{CH}-\text{X}$	Vinyl halide
	Aryl halide
	Benzylic halide

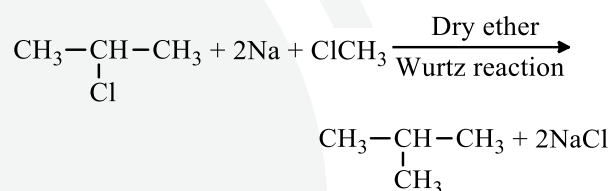
(NCERT Ed. 2025-26 12th Part-II Page No. 160)

87. (3)



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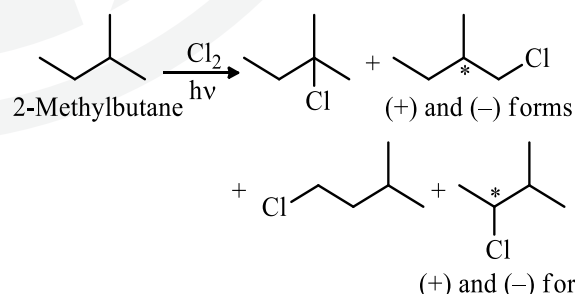
88. (3)



(NCERT Ed. 2025-26 12th Part-II Page No. 181)

89. (2)

The number of monohalogenation products obtained from any alkane depends upon the number of different types of hydrogen it contains. The possible monochlorinated products of 2-methyl butane are:



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90. (2)

2,3-Dichlorobutane have meso-isomer due to the presence of the plane of symmetry.

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91. (2)

Anatomy of Dicot root: The outermost layer is epiblema. Many of the cells of epiblema protrude in the form of unicellular root hairs. The cortex consists of several layers of thin-walled parenchyma cells with intercellular spaces. The innermost layer of the cortex is called endodermis. It comprises a single layer of barrel-shaped cells without any intercellular spaces. The pith is small or inconspicuous. There are usually two to four xylem and phloem patches.

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92. (3)

In some plants such as *Rhizophora* growing in swampy areas, many roots come out of the ground and grow vertically upwards. Such roots, called pneumatophores, help to get oxygen for respiration.

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93. (2)

The simple permanent tissue in which the cells are usually dead at maturity is sclerenchyma. Sclerenchyma consists of long, narrow cells with thick and lignified cell walls having a few or numerous pits. They are usually dead and without protoplasts. On the basis of variation in form, structure, origin and development, sclerenchyma may be either fibres or sclereids. The cells of collenchyma provide mechanical support to the growing parts of the plant such as young stem and petiole of a leaf.

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94. (1)

In pea and bean flowers, there are five petals, the largest (standard) overlaps the two lateral petals (wings) (free) which in turn overlap the two smallest anterior petals (keel) (fused); this type of aestivation is known as vexillary or papilionaceous.



Types of aestivation in corolla : Vexillary

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95. (4)

The meristems which occur at the tips of roots and shoots and produce primary tissues are called apical meristems. The meristem which occurs between mature tissues is known as intercalary meristem. They occur in grasses and regenerate parts removed by the grazing herbivores. Both apical meristems and intercalary meristems are primary meristems because they appear early in life of a plant and contribute to the formation of the primary plant body. The meristem that occurs in the mature regions of roots and shoots of many plants, particularly those that produce woody axis and appear later than primary meristem is called the secondary or lateral meristem. They are cylindrical meristems. Fascicular vascular cambium, interfascicular cambium and cork-cambium are examples of lateral meristems. These are responsible for producing the secondary tissues. Apical meristems do not disappear during secondary growth. They continue to function at the growing tips, while secondary growth (increase in girth) is carried out by lateral meristems (vascular cambium and cork cambium).

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96. (2)

In parietal placentation, the ovules develop on the inner wall of the ovary or on peripheral part. Ovary is one-chambered but it becomes two-chambered due to the formation of the false septum, e.g., mustard and *Argemone*. The label 'X' refers to the ovule that becomes seed after fertilisation. In tomato and lemon, axile placentation is seen. The placenta develops at the base of ovary in basal placentation.

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97. (3)

The outside of the epidermis is often covered with a waxy thick layer called the cuticle which prevents the loss of water. Cuticle is absent in roots.

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98. (3)

A flower having either only stamens or only carpels is unisexual. When it can be divided into two similar halves only in one particular vertical plane, it is zygomorphic, e.g., pea, gulmohur, bean, *Cassia*. When a flower can be divided into two

equal radial halves in any radial plane passing through the centre, it is said to be actinomorphic, e.g., mustard, *Datura*, chilli. A flower is asymmetric (irregular) if it cannot be divided into two similar halves by any vertical plane passing through the centre, as in canna. Flowers with bracts-reduced leaf found at the base of the pedicel - are called bracteate and those without bracts, ebracteate. In some flowers like lily, the calyx and corolla are not distinct and are termed as perianth.

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99. (2)

Dicot root: The outermost layer is epiblema. The cortex consists of several layers. The innermost layer of the cortex is called endodermis. Next to endodermis lies a few layers of thick-walled parenchymatous cells referred to as pericycle. All tissues on the innerside of the endodermis such as pericycle, vascular bundles and pith constitute the stele.

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100. (3)

The given floral formula belongs to family Malvaceae, commonly called the cotton family. An additional whorl of 3-9 leafy structures present outside calyx, called epicalyx (hypocalyx) is represented as Epik. The petals are free. The ovary is superior.

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101. (1)

The description of the diverse forms of life on earth was made only by observation – through naked eyes or later through magnifying lenses and microscopes. This description is mainly of gross structural features, both external and internal. Before experimental biology or more specifically, physiology, was established as a part of biology, naturalists described only biology. Hence, biology remained as a natural history for a long time. While the initial reaction of a student could be boredom, one should keep in mind that the detailed description, was utilised in the later day reductionist biology where living processes drew more attention from scientists than the description of life forms and their structure.

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102. (4)

Parenchyma forms the major component within organs. The collenchyma occurs in layers below the epidermis in most of the dicotyledonous plants. Xylem is composed of four different kinds of elements, namely, tracheids, vessels, xylem fibres and xylem parenchyma (the only living xylem tissue cell type). Katherine Esau reported in her early publications that the curly top virus spreads through a plant via the food-conducting or phloem tissue.

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103. (4)

In alternate type of phyllotaxy, a single leaf arises at each node in alternate manner, as in china rose, mustard and sun flower plants. In opposite type, a pair of leaves arise at each node and lie opposite to each other as in *Calotropis* and guava plants. If more than two leaves arise at a node and form a whorl, it is called whorled, as in *Alstonia*. *Alstonia* exhibits whorled phyllotaxy (more than two leaves at a node) and has simple leaves, not palmately compound leaves. In a pinnately compound leaf a number of leaflets are present on a common axis, the rachis, as in neem. In palmately compound leaves, the leaflets are attached at a common point, as in silk cotton.

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104. (1)

The monocot stem has a sclerenchymatous hypodermis, a large number of scattered vascular bundles that are conjoint and closed. A large number of vascular bundles are arranged in a ring ; the 'ring' arrangement of vascular bundles is a characteristic of dicot stem. Each vascular bundle is conjoint, open, and with endarch protoxylem in them.

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105. (2)

When the placenta is axial and the ovules are attached to it in a multilocular ovary, the placentaion is said to be axile, as in china rose, tomato and lemon.

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106. (2)

In the monocot stem, peripheral vascular bundles are generally smaller than the centrally located ones. The phloem parenchyma is absent, and water-containing cavities are present within the vascular bundles.

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107. (3)

In some seeds such as castor the endosperm formed as a result of double fertilisation, is a food storing tissue and called endospermic seeds.

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108. (3)

In stems, the protoxylem lies towards the centre (pith) and the metaxylem lies towards the periphery of the organ. This type of primary xylem is called endarch.

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109. (2)

Monocot plants characteristically possess a fibrous root system, where a cluster of roots arises from the base of the stem. Tap roots are typical of dicot plants.

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110. (4)

The monocot stem has a sclerenchymatous hypodermis, a large number of scattered vascular bundles, each surrounded by a sclerenchymatous bundle sheath, and a large, conspicuous parenchymatous ground tissue. In dorsiventral leaf, the vascular bundles are surrounded by a layer of thick walled bundle sheath cells. Isobilateral leaves (monocot leaves) do have vascular bundles surrounded by bundle sheath cells, which can be parenchymatous or sclerenchymatous.

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111. (2)

A leaf is said to be simple, when its lamina is entire or when incised, the incisions do not touch the midrib. When the incisions of the lamina reach up to the midrib breaking it into a number of leaflets, the leaf is called compound. A bud is present in the axil of petiole in both simple and compound leaves, but not in the axil of leaflets of the compound leaf.

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112. (2)

Parenchyma cells are primarily involved in storage, photosynthesis, and secretion. Their main role is not mechanical support, especially in mature organs where sclerenchyma and collenchyma take over this function. Collenchyma provide mechanical support to the growing parts of the plant such as young stem and petiole of a leaf. Sclerenchyma provides mechanical support to organs. Leaf and stem may be modified as tendrils

to provide mechanical support for climbing. Thorns/spines are modified structures for defence.

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113. (1)

Underground stems of potato, ginger, turmeric, zaminkand, *Colocasia* are modified to store food in them. A lateral branch with short internodes and each node bearing a rosette of leaves and a tuft of roots is found in aquatic plants like *Pistia* and *Eichhornia*. In some plants such as Australian acacia, the leaves are small and short-lived. The petioles in these plants expand, become green and synthesise food. Tap roots of carrot, turnip and adventitious roots of sweet potato, get swollen and store food.

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114. (4)

The diagram given in the question is a representation of stomata with bean-shaped guard cells (D). This shape of guard cells is typical for dorsiventral leaves. B, D and E together constitute the stomatal apparatus. E is the stomatal aperture or pore. Sometimes, a few epidermal cells, in the vicinity of the guard cells become specialised in their shape and size and are known as subsidiary cells (A).



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115. (3)

If gynoecium is situated in the centre and other parts of the flower are located on the rim of the thalamus almost at the same level, it is called perigynous. The ovary here is said to be half-inferior, e.g., plum, rose, peach.

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116. (4)

On the stem the epidermal hairs are called trichomes. The trichomes in the shoot system are usually multicellular. They may be branched or unbranched and soft or stiff. They may even be secretory. The trichomes help in preventing water loss due to transpiration.

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117. (2)

The correct sequence of structures in a floral diagram from the innermost to the outermost whorl is: Gynoecium (innermost, female reproductive part); Androecium (male reproductive part, surrounds the gynoecium); Corolla (petals, surrounds the androecium); Calyx (outermost, sepals).

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118. (1)

Medullary rays are primarily composed of parenchyma (simple permanent tissue) cells that extend radially in between the vascular bundles. Their main function is the radial conduction of water, minerals, and organic substances, as well as storage. Being parenchymatous, they are living cells.

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119. (4)

In majority of the dicotyledonous plants, the direct elongation of the radicle leads to the formation of primary root which grows inside the soil. It bears lateral roots of several orders that are referred to as secondary, tertiary, etc. roots. The primary roots and its branches constitute the tap root system, as seen in the mustard plant. In some plants, like grass, *Monstera* and the banyan tree, roots arise from parts of the plant other than the radicle and are called adventitious roots. The main functions of the root system are absorption of water and minerals from the soil, providing a proper anchorage to the plant parts, storing reserve food material and synthesis of plant growth regulators. The hanging structures that support a banyan tree are called prop roots.

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120. (3)

Dorsiventral leaf: Mesophyll, which possesses chloroplasts and carry out photosynthesis, is made up of parenchyma. It has two types of cells – the palisade parenchyma and the spongy parenchyma. The adaxially placed palisade parenchyma is made up of elongated cells, which are arranged vertically and parallel to each other. The oval or round and loosely arranged spongy parenchyma is situated below the palisade cells and extends to the lower epidermis. There are numerous large spaces and air cavities between these cells.

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121. (1)

The fruit in members of family Leguminosae is a legume. Cereals (family Gramineae) are one seeded fruit called caryopsis.

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122. (2)

The innermost layer of the cortex in a dicot stem is called the endodermis. The cells of the endodermis are rich in starch grains and the layer is also referred to as the starch sheath. Pericycle is present on the inner side of the endodermis and above the phloem in the form of semi-lunar patches of sclerenchyma.

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123. (3)

The fruit is a characteristic feature of the flowering plants. It is a mature or ripened ovary, developed after fertilisation. If a fruit is formed without fertilisation of the ovary, it is called a parthenocarpic fruit. In mango and coconut, the fruit is known as a drupe. They develop from monocarpellary superior ovaries and are one seeded. In mango the pericarp is well differentiated into an outer thin epicarp, a middle fleshy edible mesocarp and an inner stony hard endocarp.

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124. (4)

The tangential as well as radial walls of the endodermal cells in roots have a deposition of water-impermeable, waxy material suberin in the form of casparian strips. In grasses, certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These are called bulliform cells. The root hairs are unicellular elongations of the epidermal cells and help absorb water and minerals from the soil. Stomata regulate the process of transpiration and gaseous exchange.

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125. (2)

Ray florets have inferior ovary.

(NCERT Ed. 2025-26, 11th, Page No. 63)

126. (1)

Dorsiventral leaf: The vertical section of a dorsiventral leaf through the lamina shows three main parts, namely, epidermis, mesophyll and vascular system. The epidermis which covers both the upper surface (adaxial epidermis) and lower

surface (abaxial epidermis) of the leaf has a conspicuous cuticle. The abaxial epidermis generally bears more stomata than the adaxial epidermis. The latter may even lack stomata. Vascular system includes vascular bundles, which can be seen in the veins and the midrib. The size of the vascular bundles are dependent on the size of the veins. The veins vary in thickness in the reticulate venation of the dicot leaves.

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127. (3)

Katherine Esau's *Plant Anatomy* published in 1954 took a dynamic, developmental approach designed to enhance one's understanding of plant structure and an enormous impact worldwide, literally bringing about a revival of the discipline. The *Anatomy of Seed Plants* by Katherine Esau was published in 1960. It was referred to as Webster's of plant biology – it is encyclopaedic. In 1957 she was elected to the National Academy of Sciences, becoming the sixth woman to receive that honour. In addition to this prestigious award, she received the National Medal of Science from President George Bush in 1989.

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128. (1)

The stamens may be united into one bunch or one bundle (monadelphous) as in china rose, or two bundles (diadelphous) as in pea, or into more than two bundles (polyadelphous) as in citrus. Axillary buds of stems may also get modified into woody, straight and pointed thorns. Thorns are found in many plants such as *Citrus*, *Bougainvillea*. Some plants of arid regions modify their stems into flattened (*Opuntia*), or fleshy cylindrical (*Euphorbia*) structures. There may be a variation in the length of filaments within a flower, as in *Salvia* and mustard.

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129. (1)

The anatomy of the monocot root is similar to the dicot root in many respects. It has epidermis, cortex, endodermis, pericycle, vascular bundles and pith. There are usually more than six (polyarch) xylem bundles in the monocot root. Pith is large and well developed. Monocotyledonous roots do not undergo any secondary growth. The vascular bundles are said to be closed.

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130. (1)

In racemose type of inflorescences the main axis continues to grow, the flowers are borne laterally in an acropetal succession. In cymose type of inflorescence the main axis terminates in a flower, hence is limited in growth. The flowers are borne in a basipetal order.

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131. (3)

Phloem fibres (bast fibres) are made up of sclerenchymatous cells. These are generally absent in the primary phloem but are found in the secondary phloem. These are much elongated, unbranched and have pointed, needle like apices. The cell wall of phloem fibres is quite thick. At maturity, these fibres lose their protoplasm and become dead.

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132. (4)

When more than one carpel is present, they may be free (as in lotus and rose) and are called apocarpous. When stamens are attached to the petals, they are epipetalous as in brinjal, or epiphyllous when attached to the perianth as in the flowers of lily. Corolla is said to be gamopetalous when petals are united as in tomato/potato.

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133. (2)

Phloem parenchyma is absent in most of the monocotyledons.

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134. (1)

A typical flower has four different kinds of whorls arranged successively on the swollen end of the stalk or pedicel, called thalamus or receptacle.

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135. (3)

The outer walls of guard cells (away from the stomatal pore) are thin and the inner walls (towards the stomatal pore) are highly thickened. On the basis of their structure and location, there are three types of tissue systems. These are the epidermal tissue system, the ground or fundamental tissue system and the vascular or conducting tissue system. Gymnosperms have albuminous cells and sieve cells. They lack sieve tubes and companion cells.

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136. (3)

Scoliodon (Dogfish) belongs to the Class Chondrichthyes (Cartilaginous fishes). A key characteristic of this class is that the notochord is persistent throughout their life.

Balaenoptera (Whale) and *Delphinus* (Dolphin) are mammals, while *Aptenodytes* (Penguin) is a bird. In these animals (higher vertebrates), the notochord is an embryonic structure that is replaced by a vertebral column (backbone) in the adult stage.

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137. (1)

Earthworm: Belongs to the Phylum Annelida. A defining characteristic of Annelids is true metameric segmentation, where the body is divided externally by grooves and internally by septa into a series of repeated segments (metameres) containing serial repetition of organs (like nephridia, muscles, and nerve ganglia).

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138. (3)

Platyhelminthes (flatworms) are acoelomate animals, meaning they lack a body cavity (coelom) and they exhibit organ-level organization for the first time in the animal kingdom.

In parasitic flatworms like *Fasciola* (liver fluke), fertilization is internal, and development involves multiple larval stages.

Flame cells (also known as protonephridia) are excretory and osmoregulatory structures, they help in removal of metabolic waste and regulation of water balance.

Planaria, a free-living flatworm, shows an exceptional capacity for regeneration, being able to regenerate an entire body from small fragments.

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139. (1)

While forelimbs are modified into wings in all birds, not all birds can fly, e.g., ostrich, emu, cassowary, kiwi, and penguin are flightless.

Many long bones in birds contain air cavities (pneumatic bones) connected to air sacs. This adaptation reduces body weight and helps in efficient respiration during flight.

The crop stores food temporarily, while the gizzard mechanically grinds it. Both are specializations in the avian digestive system.

Birds have dry, non-glandular skin covered by feathers. They possess one oil gland, the uropygial gland, near the tail, used for feather maintenance.

Birds have lungs assisted by air sacs, allowing continuous unidirectional airflow, which makes their respiration highly efficient.

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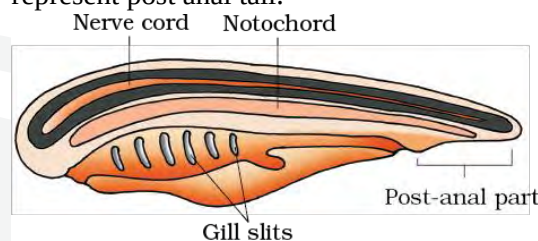
140. (4)

List-I	List-II
Annelids	Possess longitudinal and circular muscles
Platyhelminthes	Flatworms
Aschelminthes	Polyp and medusa
Coelenterata	Roundworms

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141. (2)

In the given figure 'A' represent nerve cord, 'B' represent notochord, 'C' represent gill slits and 'D' represent post anal tail.



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142. (3)

Arthropods dominate most habitats, and their success is largely due to a chitinous exoskeleton (protection, prevention of desiccation, muscle attachment), jointed appendages (specialized locomotion/feeding/sensory roles), and—in terrestrial groups like insects—a tracheal system that supplies oxygen directly to tissues, supporting high activity levels.

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143. (4)

Ctenophores, commonly known as sea walnuts or comb jellies are exclusively marine, radially symmetrical, diploblastic organisms with tissue level of organisation. The body bears eight external rows of ciliated comb plates, which help in locomotion. Digestion is both extracellular and intracellular. Reproduction takes place only by sexual means.

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144. (1)

Phylum Chordata is divided into three subphyla: Urochordata or Tunicata, Cephalochordata and Vertebrata. In Urochordata, notochord is present only in larval tail.

Lancelet (*Amphioxus/Branchiostoma*) is Cephalochordate.

Chordates have a dorsal, hollow central nervous system.

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145. (3)

The figure is of a tapeworm (*Taenia*), belongs to phylum platyhelminthes.

Tapeworms are triploblastic acoelomates.

The “segments” are pseudosegmentation, not true metamerism.

They lack a digestive tract and absorb nutrients directly through the body surface.

High regeneration is characteristic of planarians.

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146. (1)

Platyhelminthes (flatworms) usually show internal fertilisation. Most are hermaphrodites, so male and female gametes are produced within the same individual.

The body of platyhelminthes are flattened dorsoventrally, helping with diffusion as they lack a circulatory system.

Many platyhelminths are endoparasitic, living inside the host’s body.

Planarians, a free-living group of flatworms, are famous for their remarkable regeneration capacity.

Most Platyhelminthes are hermaphroditic (monoecious), having both male and female reproductive organs in the same individual.

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147. (1)

Species A: one plane through the longitudinal axis → Bilateral symmetry (e.g., annelid).

Species B: many planes through a central axis → Radial symmetry (classic examples: cnidarians like *Hydra*/jellyfish; adult echinoderms).

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148. (4)

Animals belonging to Annelida, Arthropoda and Mollusca phyla have true coelom and bilateral symmetry. Coelenterates and Ctenophores are acoelomates and show radial symmetry. Platyhelminthes are acoelomates and show bilateral symmetry. Aschelminthes are pseudocoelomates and show bilateral symmetry.

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149. (3)

Aplysia belongs to phylum Mollusca. *Gorgonia*, *Hydra*, *Obelia*, *Pennatul*a, *Adamsia*, *Physalia* belongs to phylum Cnidaria.

Spongilla and *Scypha* belongs to phylum Porifera, *Echinus* belongs to phylum Echinodermata.

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150. (1)

Roundworms are pseudocoelomates because their body cavity is not lined by mesoderm, instead, the mesoderm is present as scattered pouches in between the ectoderm and endoderm.

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151. (4)

Cnidarians have true tissues and a nerve net, enabling coordinated movement vs. cellular-level sponges.

Platyhelminthes show organ level organization (e.g., protonephridia with flame cells; eyespots).

Aschelminthes have organ-system level with a complete digestive tract.

Arthropods possess a highly advanced organ-system level; “cells loosely arranged in aggregates” describes Porifera.

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152. (1)

Embryo X: Two germ layers with non-cellular matrix (mesoglea) → Diploblastic, Acoelomate (e.g., Cnidaria).

Embryo Y: Three germ layers; cavity not lined by mesoderm (mesoderm as scattered pouches) → Triploblastic, Pseudocoelomate (e.g., Nematoda).

Embryo Z: Three germ layers; cavity fully lined by mesoderm → Triploblastic, Eucoelomate (e.g., Annelida/Mollusca).

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153. (2)

List-I	List-II
Fighting fish	<i>Betta</i>
Angel fish	<i>Pterophyllum</i>
Flying fish	<i>Exocoetus</i>
Saw fish	<i>Pristis</i>

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154. (4)

Pavo is the genus of peacock (e.g., *Pavo cristatus*). Penguin belongs to genera like *Aptenodytes* (emperor/king penguins) or *Spheniscus* (banded penguins).

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155. (1)

Aschelminthes include many free-living forms and plant parasites; they aren’t exclusively parasitic in animals.

Female roundworms are generally larger/longer than males (common sexual dimorphism; e.g., *Ascaris*).

Aschelminthes are pseudocoelomates with a complete digestive tract, organizationally more advanced than acoelomate Platyhelminthes.

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156. (1)

List-I (Level of organisation)	List-II (Animal)
Organ level	<i>Fasciola</i>
Cellular aggregate level	<i>Spongilla</i>
Tissue level	<i>Obelia</i>
Organ system level	<i>Pheretima</i>

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157. (2)

Ancylostoma duodenale is a nematode and are pseudocoelomate.

Sexes are separate (dioecious); males are smaller/shorter than females.

It is parasitic in the human small intestine.

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158. (3)

In Annelids (e.g., earthworm), blood flows through closed vessels → closed circulatory system.

In Arthropods (e.g., cockroach, prawn), blood is not confined to vessels but flows into hemocoel → open circulatory system.

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159. (3)

Sponges rely on a continuous water current that enters through ostia and exits through the osculum. This current carries both oxygen and microscopic food particles to the cells.

Without the flow of nutrient-rich water, their metabolic activity slows down or stops, even if oxygen is available.

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160. (4)

Reptiles are poikilothermic (ectothermic); their heart is generally three-chambered (except crocodilians, which are four-chambered).

Snakes and many lizards shed the skin along with the epidermal scales.

Not all reptiles have external ear openings; snakes lack a tympanum/external ear opening.

Reptiles are predominantly oviparous and show direct development (no larval stage).

When present, reptilian limbs occur as two pairs (snakes are limbless).

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161. (3)

List-I	List-II
Ostia	Tiny pores in the body through which water enters
Osculum	Single, large excurrent opening
Spongocoel	Central cavity lined by choanocytes
Choanocytes	Flagellated collar cells that generate water current and trap food particles

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162. (4)

Naja - Cobra

Bangarus - Krait

Vipera - Viper

Hemidactylus - common house/wall lizard

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163. (2)

Frogs have a three-chambered heart with two atria (auricles) and one ventricle.

Frogs have a closed circulatory system (blood remains within vessels) and are cold-blooded, meaning their body temperature varies with the environment.

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164. (1)

Balaenoptera → Blue whale genus — a fully aquatic mammal.

Delphinus → Dolphin — A fully aquatic mammal.

Equus → Horse — terrestrial.

Pteropus → Flying fox (bat) — aerial.

Felis → Cat — terrestrial.

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165. (2)

Class Osteichthyes includes both marine and fresh water fishes with bony endoskeleton. Their body is streamlined. Mouth is mostly terminal

Amphibians (e.g., frogs and salamanders) have moist, glandular skin without scales, which helps in cutaneous respiration.

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166. (4)

Examples of molluscs: *Pila* (Apple snail), *Pinctada* (Pearl oyster), *Sepia* (Cuttlefish), *Loligo* (Squid), *Octopus* (Devil fish), *Aplysia* (Seahare), *Dentalium* (Tusk shell) and *Chaetopleura* (Chiton).

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167. (2)

Pristis (sawfish) is a marine cartilaginous fish, but Chondrichthyes have a cartilaginous endoskeleton, not a “cartilaginous exoskeleton.” (Their external covering is placoid scales, not cartilage.)

In cartilaginous fishes like *Pristis*, the mouth is ventral and the notochord persists throughout life (partly enclosed by cartilaginous vertebrae).

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168. (3)

Pheretima (earthworm) excretes via nephridia.

Saccoglossus (a hemichordate) excretes mainly via a proboscis gland (glomerulus) located in the proboscis.

Malpighian tubules → arthropods;

protonephridia → flatworms/larval forms;

green glands → crustaceans.

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169. (3)

All of these animals (silverfish, scorpion, dragonfly, prawn) belong to Phylum Arthropoda.

Common features of Arthropods: Jointed appendages, chitinous exoskeleton

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170. (4)

Apis or honey bee belongs to phylum Arthropoda has body consists of head, thorax and abdomen. It has jointed appendages. Tracheal system is present for respiration. Circulatory system is open type. It has economic importance as it produces honey, bee-wax and other useful bee products.

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171. (2)

The most unique mammalian characteristic is the presence of milk producing glands or mammary glands.

Air sacs connected to lungs supplement respiration in birds.

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172. (2)

List-I	List-II
Locust	A gregarious pest causing crop damage
Butterfly	Important pollinator for flowering plants
Prawn	Economically important as a food source
Mosquito	Vector for diseases like dengue and chikungunya

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173. (2)

Pavo belongs to class Aves, their digestive tract has additional chambers, the crop and gizzard. *Periplaneta* (Cockroach) has crop and gizzard in their alimentary canal, crop helps in storage of food. *Pavo* have tendency to maintain their body temperature whereas *Periplaneta* does not have tendency to regulate their body temperature. Both are coelomate. Exoskeleton of *Periplaneta* is made of N-acetyl glucosamine whereas endoskeleton of *Pavo* is made of bone and cartilage.

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174. (3)

Petromyzon belongs to class Cyclostomata of sub phylum vertebrata and does not contains jaw, so it is a jawless vertebrate not invertebrate.

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175. (4)

List-I	List-II
<i>Macropus</i>	Hair on skin
<i>Struthio</i>	Feathers on skin
<i>Hemidactylus</i>	Dry cornified skin
<i>Rana</i>	Moist skin

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176. (1)

Bioluminescence (the ability of a living organism to emit light) is a defining and "well-marked" characteristic of the Phylum Ctenophora (Comb Jellies). Examples include *Pleurobrachia* and *Ctenoplana*.

The phenomenon of bioluminescence occurs because these animals possess specialized cells (often called photocytes) located in their meridional canals. Inside these cells, biochemical reactions (typically involving photoproteins activated by Calcium ions) take place to generate light.

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177. (3)

Birds and mammals are homeothermic (warm-blooded), not poikilothermic.

Betta and *Labeo* are bony fishes, not cartilaginous fishes.

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178. (2)

In echinodermata digestive system is well developed whereas excretory system is absent.

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179. (3)

In hemichordates, sexes are separate (dioecious), fertilisation is external and development is indirect.

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180. (4)

List-I	List-II
Aschelminthes	Often wormshaped, elongated
Annelida	Body segmentation like rings
Arthropoda	Exoskeleton of cuticle, jointed appendages
Mollusca	This is the second largest animal phylum

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